HW2-3

De morgan's Theorem

proof $(AB)' = A' + B'$

1° $(AB)(A' + B')$

$= AA'B + ABB'$

$= 0$

2° $AB + A' + B'$

$= (A' + A)(A' + B) + B'$

※ By 分配律
$X + YZ = (X + Y)(X + Z)$

$= A' + B + B'$

$= 1$

Therefore

$(AB)' = A' + B'$

※ $\begin{cases} XX' = 0 \\ X + X' = 1 \end{cases}$

HW2-4

when zx=0, nx=1, zy=0, ny=0, f=1, no=1

x　　x'　　1　　1　　x'+y　　(x'+y)'

These bits instruct how to preset the x input		These bits instruct how to preset the y input		This bit selects between + / And	This bit inst. how to postset out	Resulting ALU output
zx	nx	zy	ny	f	no	out=
if zx then x=0	if nx then x=!x	if zy then y=0	if ny then y=!y	if f then out=x+y else out=x&y	if no then out=!out	f(x,y)=
1	0	1	0	1	0	0
1	1	1	1	1	1	1
1	1	1	0	1	0	-1
0	0	1	1	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	1	!x
1	1	0	0	0	1	!y
0	0	1	1	1	1	-x
1	1	0	0	1	1	-y
0	1	1	1	1	1	x+1
1	1	0	1	1	1	y+1
0	0	1	1	1	0	x-1
1	1	0	0	1	0	y-1
0	0	0	0	1	0	x+y
0	1	0	0	1	1	x-y
0	0	0	1	1	1	y-x
0	0	0	0	0	0	x&y
0	1	0	1	0	1	x|y

proof: $(x'+y)' = x-y$

$Out = (x'+y)'$

$= -(x'+y) - 1$

$= -(-x-1+y) - 1$

$= x - y + 1 - 1$

$= x - y$

※ 2's complement

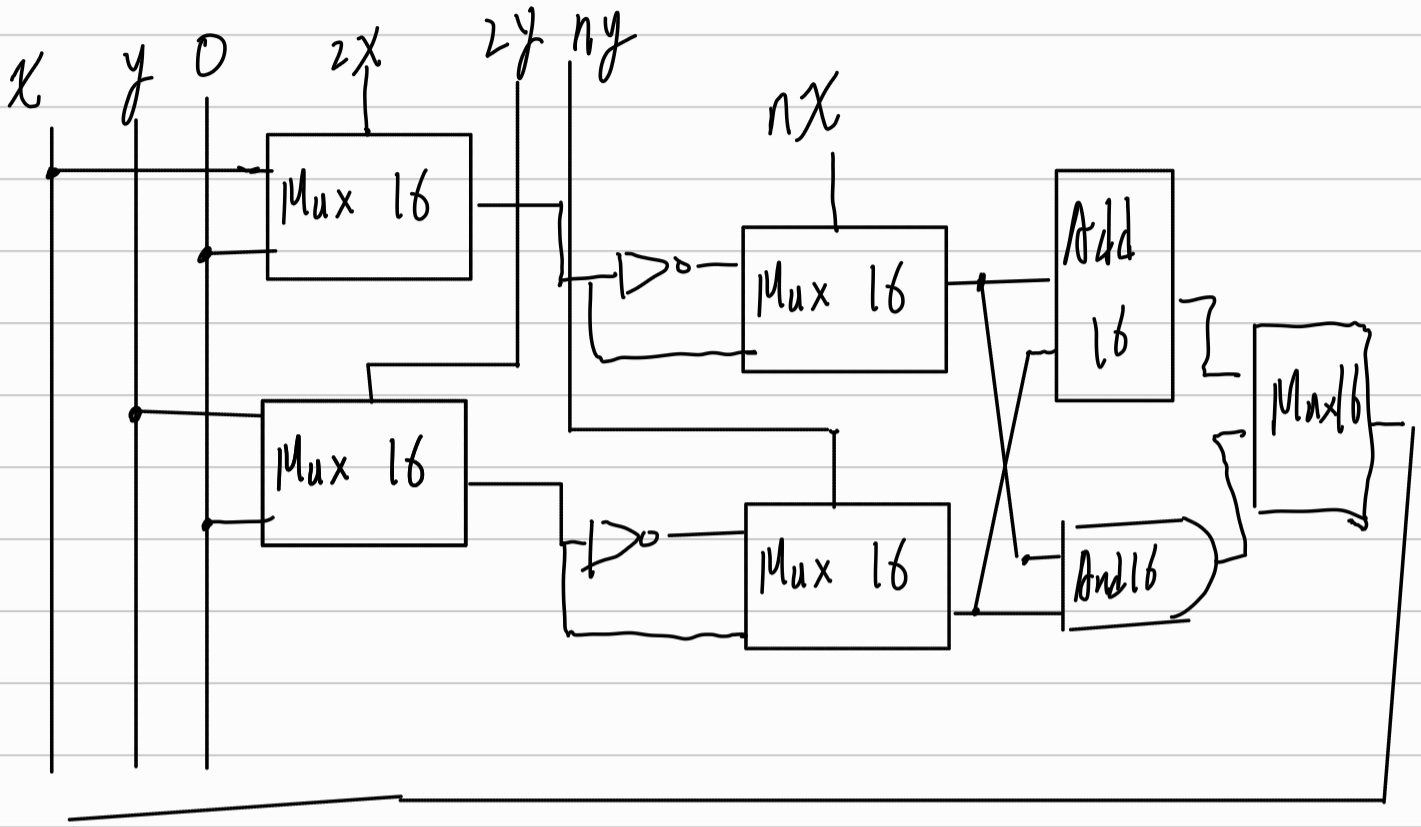
$-x = x' + 1$

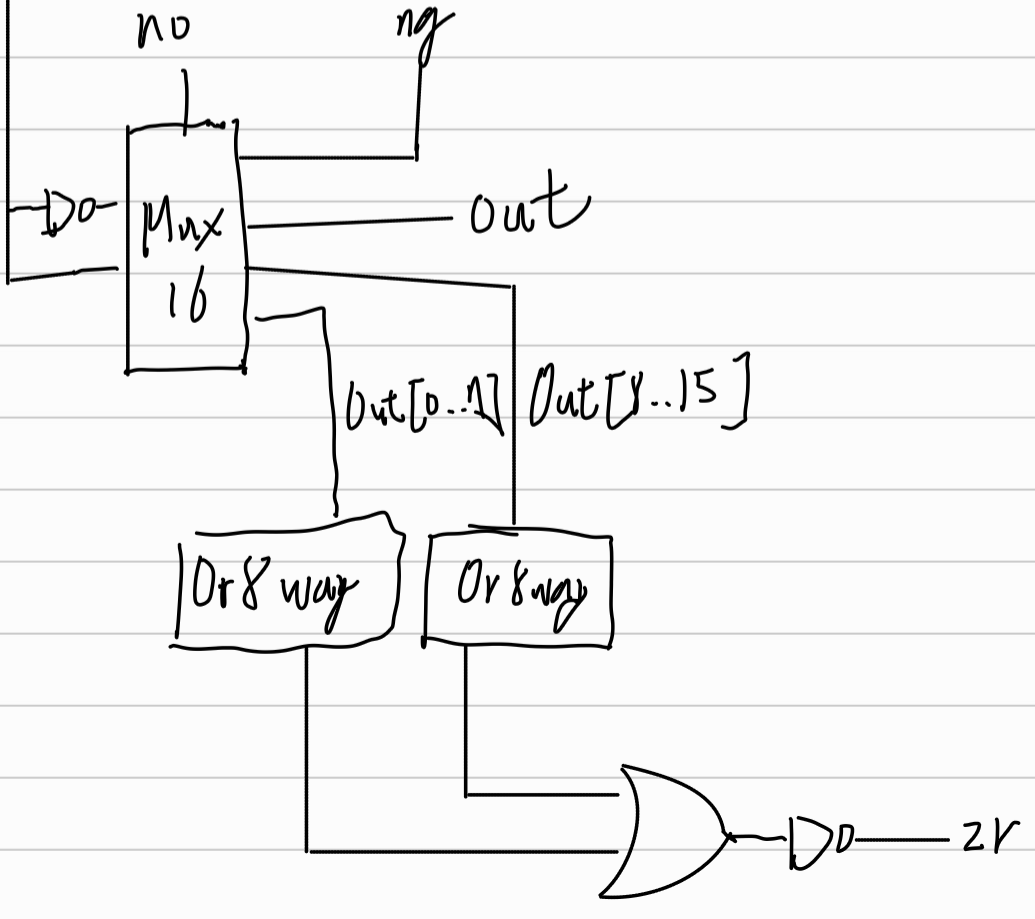
$\Rightarrow x' = -x - 1$

※這裡的 x, y 並非是布林代數的 0 or 1 而是 16 bits 的 2 進位算術，因此不能用以往 Truth table 證明

HW2-5

Draw the ALU circuit diagram.



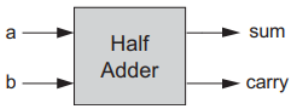


1. half adder

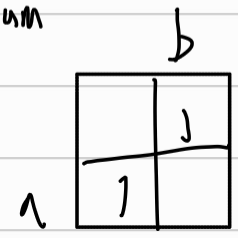
$$carry = ab = [(ab)']'$$

$$sum = ab' + a'b = (a \oplus b)$$
$$= [(ab' + a'b)']'$$
$$= [(ab)' \cdot (a'b)']'$$

Inputs		Outputs	
a	b	carry	sum
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

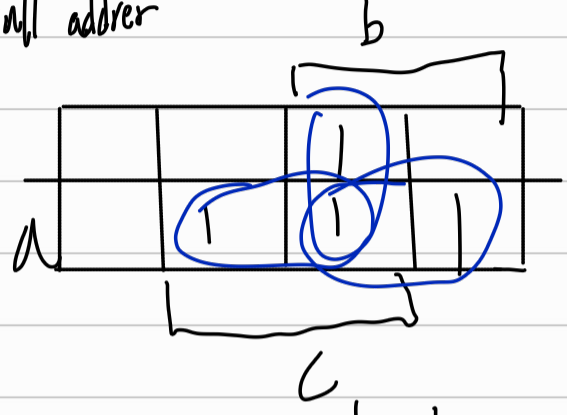


Sum

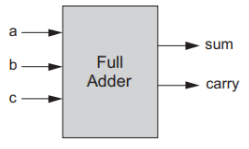


2. full adder

carry



a	b	c	carry	sum
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

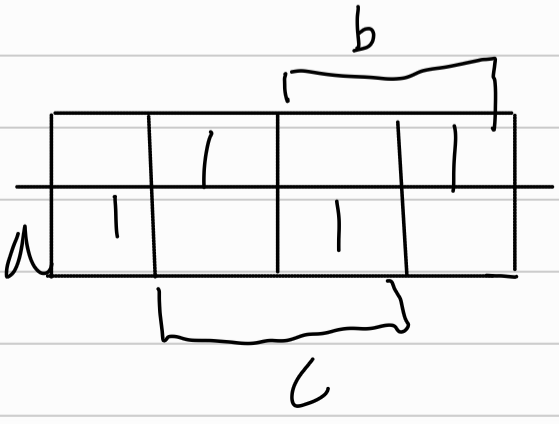


$$carry = ac + ab + bc$$
$$= a(b+c) + bc$$
$$= a[(b \oplus c) + bc] + bc$$
$$= abc + bc + a(b \oplus c)$$
$$= bc + a(b \oplus c)$$

※利用 OR and XOR 的等價

$$\Rightarrow (A+B) = (A \oplus B) + AB$$

Sum :



$Sum = ab'c' + a'bc' + abc + a'b'c$

$= c'(ab' + a'b) + c(ab + a'b')$

$= c'(a \oplus b) + c(a \oplus b)'$

令 $X = a \oplus b$

$Sum = c'X + cX'$

$= c \oplus X$

$= c \oplus a \oplus b$

※ $(A \oplus B)' = (ab' + a'b)'$

$= (ab)' \cdot (a'b)'$

$= (a' + b) \cdot (a + b')$

$= aa' + ab' + a'b + bb'$

$= ab' + a'b$

3. ALU

These bits instruct how to preset the x input		These bits instruct how to preset the y input		This bit selects between + / And	This bit inst. how to postset out	Resulting ALU output
zx	nx	zy	ny	f	no	out=
if zx then x=0	if nx then x=!x	if zy then y=0	if ny then y=!y	if f then out=x+y else out=x&y	if no then out=!out	f(x,y)=
1	0	1	0	1	0	0
1	1	1	1	1	1	1
1	1	1	0	1	0	-1
0	0	1	1	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	1	!x
1	1	0	0	0	1	!y
0	0	1	1	1	1	-x
1	1	0	0	1	1	-y
0	1	1	1	1	1	x+1
1	1	0	1	1	1	y+1
0	0	1	1	1	0	x-1
1	1	0	0	1	0	y-1
0	0	0	0	1	0	x+y
0	1	0	0	1	1	x-y
0	0	0	1	1	1	y-x
0	0	0	0	0	0	x&y
0	1	0	1	0	1	x|y